Design and Analysis of a Two Stage Miller Compensated OTA in 180nm CMOS

Abstract

This work presents the design and simulation of a two stage Operational Transconductance Amplifier (OTA). The architecture utilizes a differential input stage followed by a common source gain stage to meet high voltage gain requirements. Stability is maintained through frequency compensation, and the design's performance is validated through SPICE simulations. The results demonstrate high gain and stable operation across the desired range of frequency.

1 Project Objectives

The primary objective of this project was to design a CMOS OTA that adheres to the following performance targets:

- Achieve an open loop voltage gain exceeding 60 dB.
- Ensure frequency stability with a Phase Margin $\geq 60°$.
- Optimize bandwidth and slew rate for general purpose analog applications.
- Validate performance using 180nm TSMC process parameters.

2 Circuit Theory and Architecture

2.1 Two Stage Topology

To overcome the gain limitations of single stage amplifiers, a cascaded architecture is employed. This configuration effectively stacks the gain of two distinct stages:

1. **First Stage (Differential Pair):** Converts the differential input voltage ($V_{in+} - V_{in-}$) into a current. The voltage gain is defined by:

$$A_1 = g_{m1} \cdot r_1$$

   where g_{m1} is the transconductance of the input pair and r_1 is the output resistance of the first stage.

2. **Second Stage (Common Source):** Provides additional voltage amplification. Its gain is:

$$A_2 = g_{m6} \cdot r_2$$

   The total open loop gain is the product of the two stages:

$$A_v = A_1 \times A_2 = (g_{m1} r_1)(g_{m6} r_2)$$

2.2 Frequency Compensation

The introduction of a second gain stage adds a low frequency pole, which can degrade the phase margin. To address this, a Miller compensation capacitor C_c is connected across the second stage. This mechanism facilitates pole splitting, ensuring stability:

- The **dominant pole** (p_1) is shifted toward a lower frequency:

$$p_1 \approx \frac{1}{g_{m6} r_2 r_1 C_c}$$

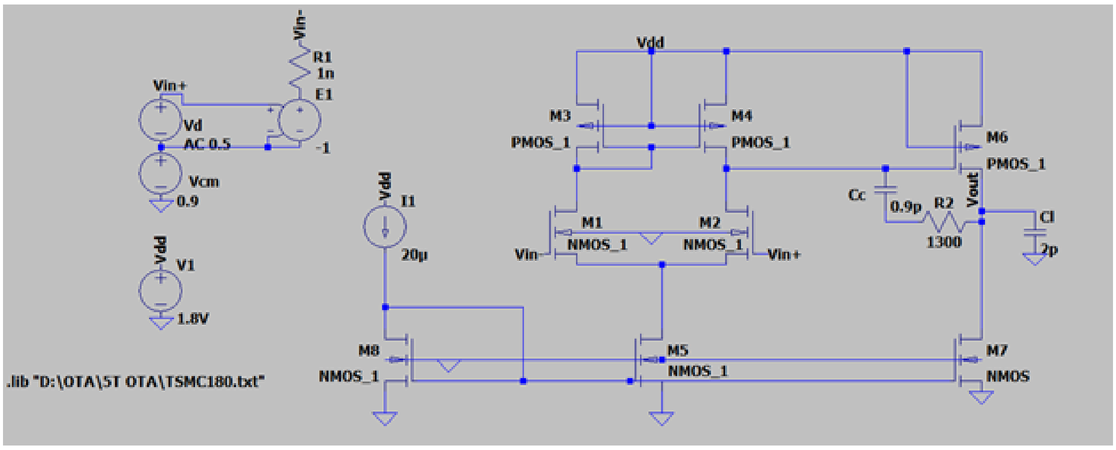
- The **non dominant pole** (p_2) is shifted toward a higher frequency:

$$p_2 \approx \frac{g_{m6}}{C_L}$$

3 Design Implementation

3.1 Schematic Configuration

The design incorporates an NMOS differential pair with PMOS active loads. The bias network and compensation circuitry are integrated as shown in the schematic below.



3.2 Transistor Sizing

Following analytical calculations and iterative tweaking to meet the specified performance metrics, the final W/L ratios were determined as follows:

Device	Type	Width (W)	Length (L)	Multiplier (M)
M_1, M_2	NMOS	7 μm	1 μm	1
M_3, M_4	PMOS	6 μm	1 μm	1
M_5	NMOS	4.35 μm	1 μm	5
M_6	PMOS	9 μm	1 μm	10
M_7	NMOS	5.5 μm	1 μm	11
M_8	NMOS	22 μm	1 μm	1

Passive Component Values:

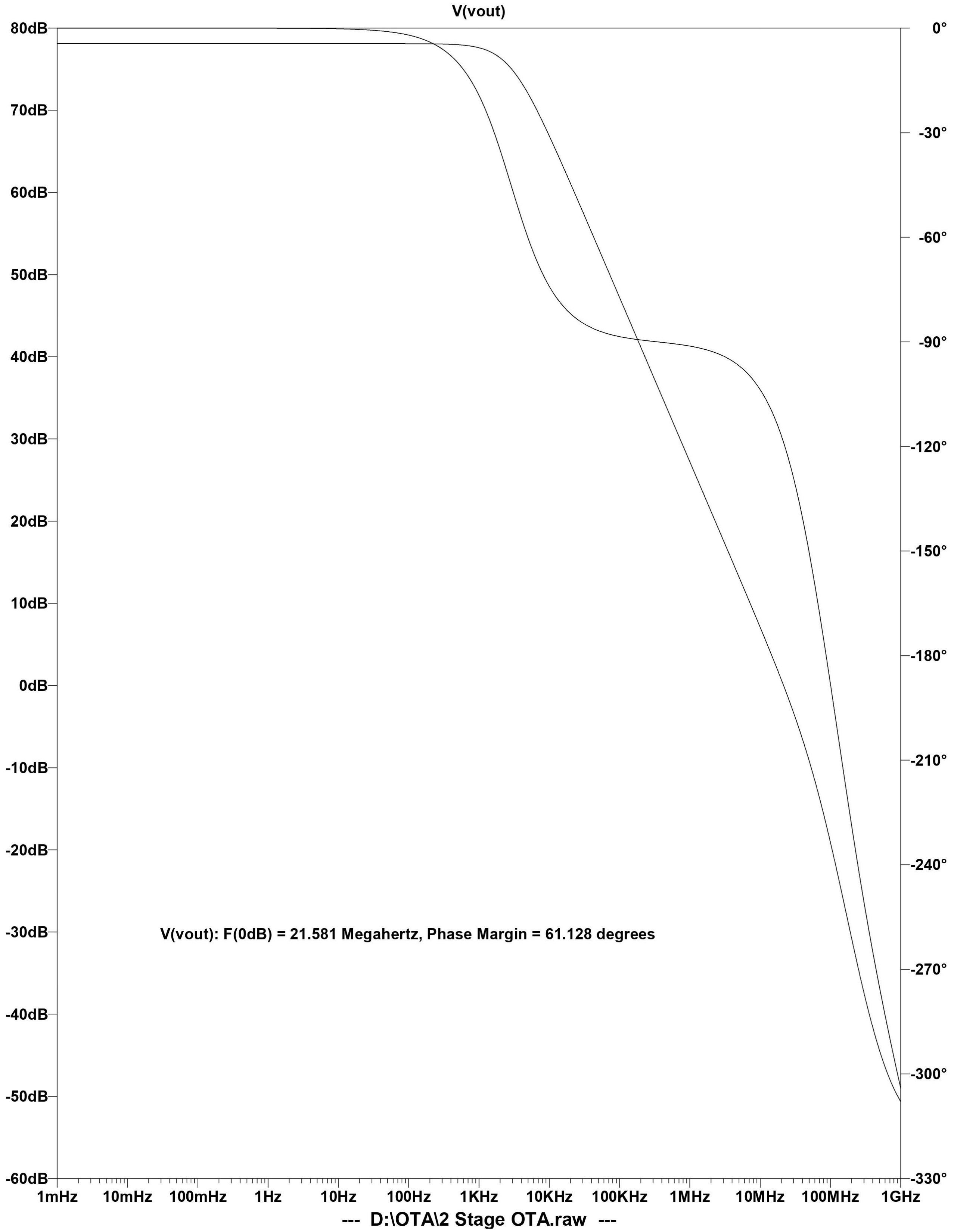
- **Compensation Capacitor (C_c):** 0.9 pF
- **Nulling Resistor (R_z):** 1.3 $k\Omega$

4 Simulation and Performance Analysis

The SPICE results show that the circuit is almost working as expected, meeting the majority of design constraints.

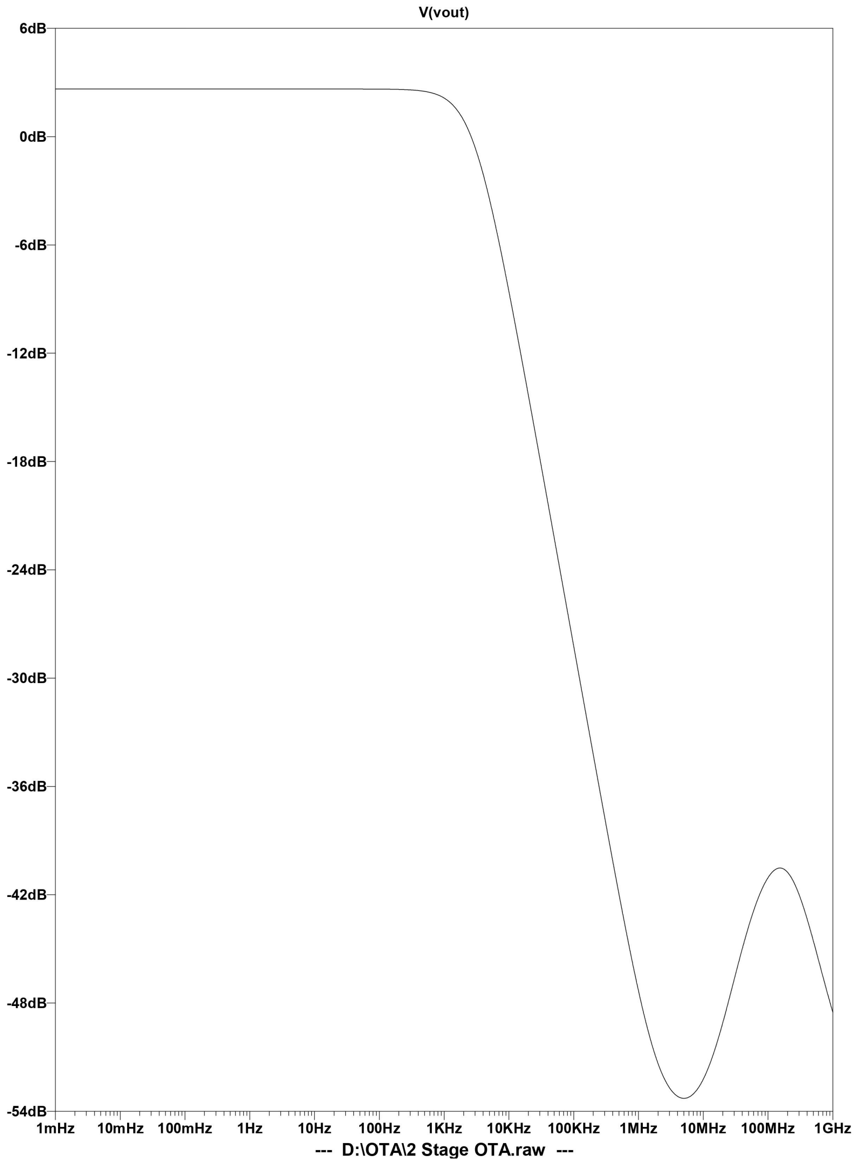
4.1 AC Analysis (Differential Mode)

The frequency response below highlights the open loop gain and the corresponding phase margin at the unity gain frequency.



4.2 Common Mode Performance

The common mode gain is analyzed to verify the amplifier's ability to reject common mode signals.



5 Summary of Results

Metric	Result
Differential Gain	77 dB
Common Mode Gain	2.6 dB
CMRR	74.4 dB
Unity Gain Bandwidth	21.5 MHz
3dB Frequency	3.135 kHz
Phase Margin	61.1°

6 Future Improvements and Optimization

While the current design satisfies the core requirements, several enhancements could be explored in future iterations to further push performance limits:

- **Enhanced Gain:** Cascoding the input differential pair or the second stage could significantly increase the DC gain.
- **Output Swing:** Utilizing rail-to-rail output stages would improve signal integrity for low-voltage applications.
- **Noise Reduction:** Implementation of flicker noise reduction techniques through device sizing or chopping could improve the signal-to-noise ratio.
- **Power Efficiency:** Adaptive biasing could be implemented to reduce quiescent power consumption without sacrificing bandwidth.

7 References

1. Behzad Razavi, *Design of Analog CMOS Integrated Circuits*.
2. Prof. Shanti Pavan, *Analog Electronics*, NPTEL.
3. Prof. Nagendra Krishnapura, *Analog Integrated Circuits*, NPTEL.